

This final diagram in the analogue computer flow diagram that simulate the second order differential equation  $a\ddot{x} + b\dot{x} + cx = 0$  which gives the solution of equation or system.

### Exercise

Simulate the analogue diagram for the following equation,

$$4\frac{d^2x}{dt^2} + \frac{3dx}{dt} + 2x = 0$$

### 5.5 Analogue computer flow diagram simulating systems

The equations and simulation of the analogue computer flow diagrams of the following systems;

1. Mechanical vibration system
2. Electrical system
3. Speed/position control system

### 5.5.1 Mechanical vibration system

A mechanical vibration system can be illustrated with the spring dashpot system shown below in fig. 5.5.1

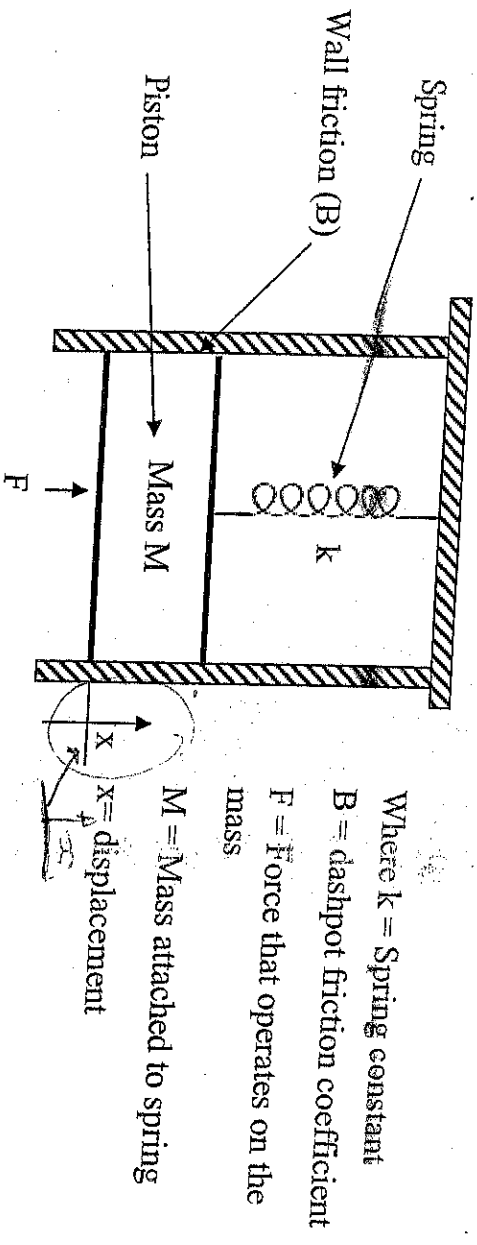


Fig. 5.5.1 Spring dashpot vibration system

The force equation for the system can be written as

$$F - B \frac{dx}{dt} - kx = M \frac{d^2x}{dt^2}$$

$$\text{Or } F = \frac{M d^2x}{dt^2} + B \frac{dx}{dt} + kx \quad (1)$$

Equation (1) is a second order differential equation whose analogue computer flow diagram can be obtained as follows;

$$\text{Equation is } F = M\ddot{x} + B\dot{x} + kx \quad (2)$$

$\ddot{x}$  must be integrated to produce  $-\dot{x}$  and  $-\dot{x}$  integrated to produce  $+x$ . The flow diagram of there operations is shown in fig. 1 below.

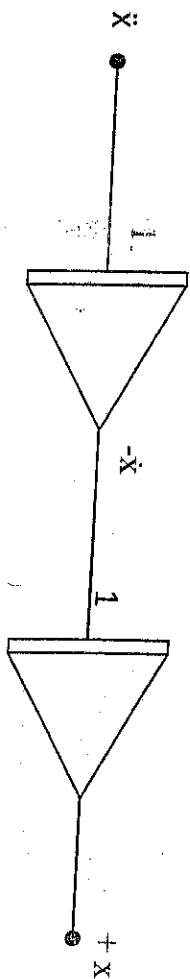


Fig. 1.0

From equation (2) solve for  $\ddot{x}$  as follows;

$$M\ddot{x} = F - B\dot{x} - kx$$

$$\therefore \ddot{x} = F/M - \frac{B}{M}\dot{x} - \frac{K}{M}x$$

This result says we can obtain  $\ddot{x}$ , from  $\dot{x}$  and  $x$  by means of a summing amplifier and three potentiometers as follows; in fig. 2

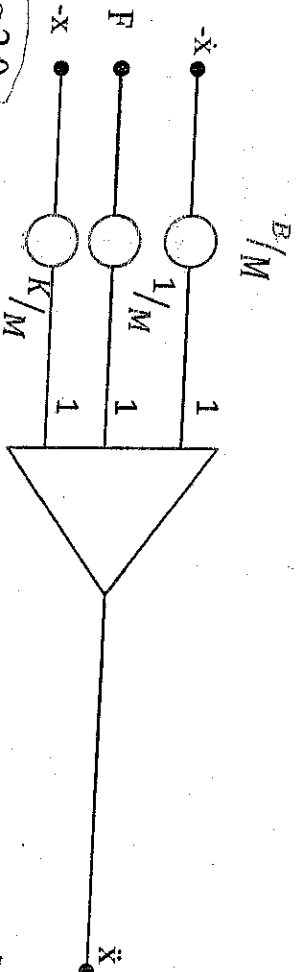


Fig 2.0

Fig. 2.0

To close the loop, we must combine fig. 1 and fig. 2 as shown below in Fig. 3.0

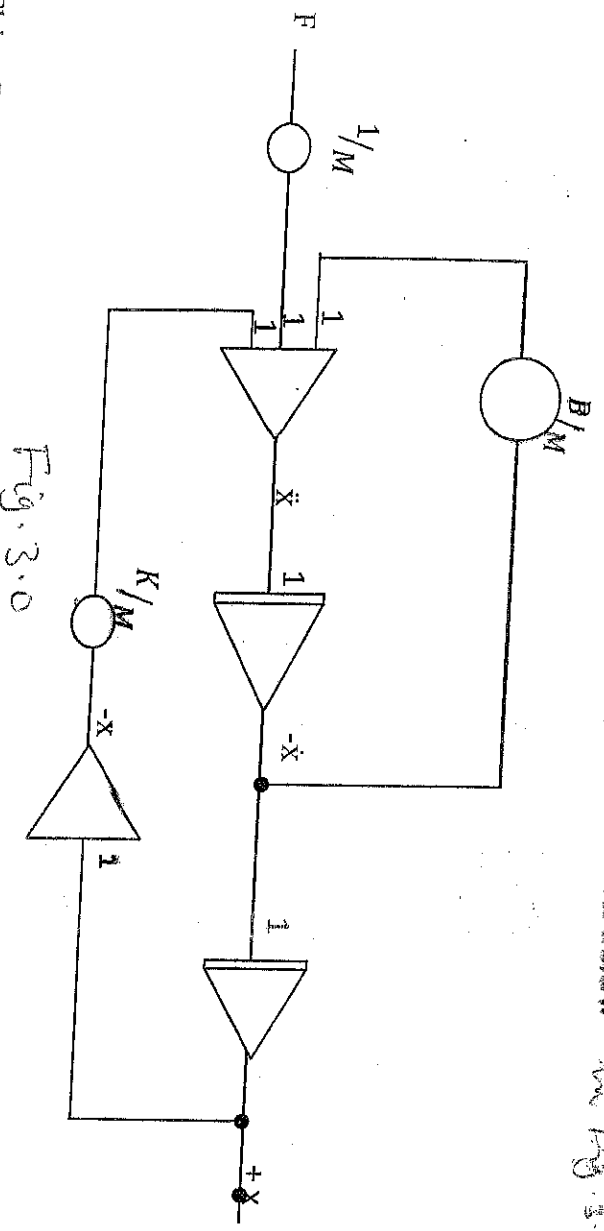


Fig. 3.0

This final diagram is the analogue computer flow diagram that simulates the equation  $F = M\ddot{x} + B\dot{x} + Kx$  which is a mechanical vibration given by spring dashpot system.

### 5.5.2 Electrical systems

An electrical system can be illustrated by R-L series circuit shown below in fig. 5.5.2

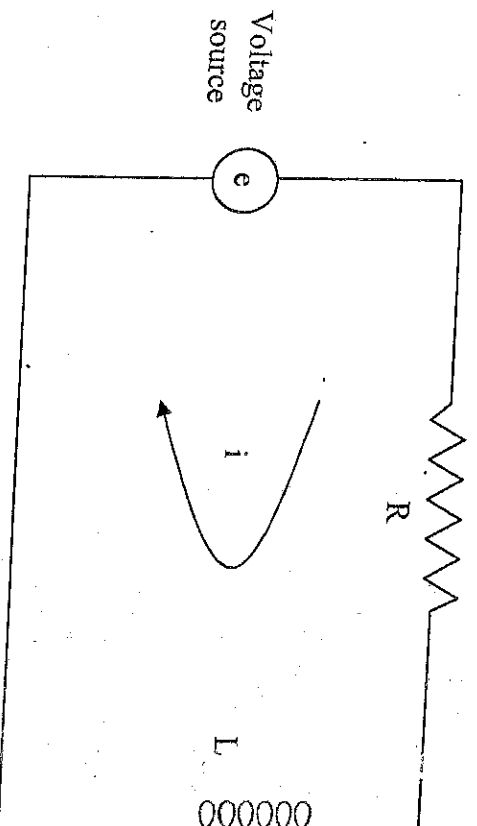


Fig. 5.5.2(a) series R-L series circuit

The equation governing the system is

$$L \frac{di}{dt} + Ri = e$$

$$\text{Or } \frac{L di}{dt} = e - Ri$$

$$\text{Or } \frac{di}{dt} = \frac{e}{L} - \frac{R}{L} i$$

The variables in this equation are  $\frac{di}{dt}$  and  $i$ . The analogue diagram that can represent  $\frac{di}{dt}$  and  $i$  are as follows;

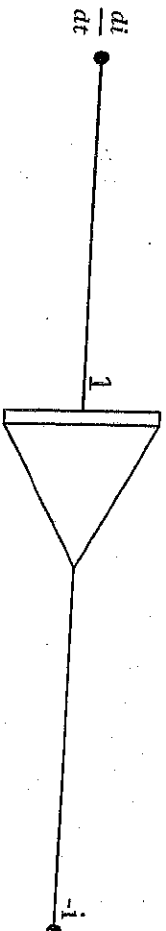


Fig. 1.0

Fig. 1.0

Adding a potentiometer to fig. 1  $\frac{R}{L} i$  can be generated as follows; in Fig. 2.0

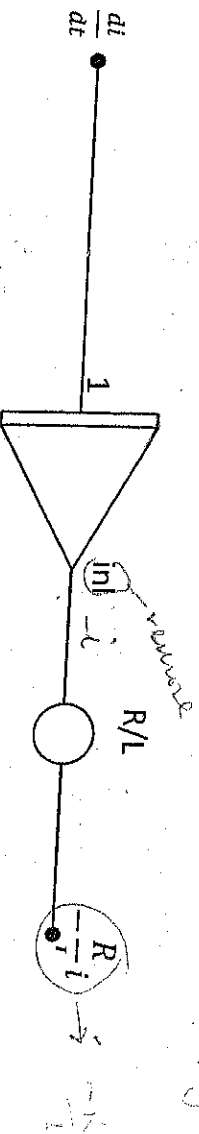


Fig. 2.0

Also,  $\frac{e}{L}$  can be generated by passing  $e$  through a potentiometer as follows

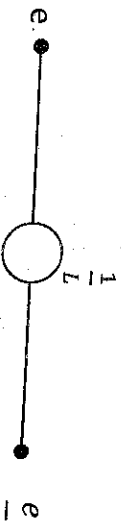


Fig. 3.0

Combining the last two diagrams and, incorporating summer gives the flow diagram below in Fig. 5.5.2

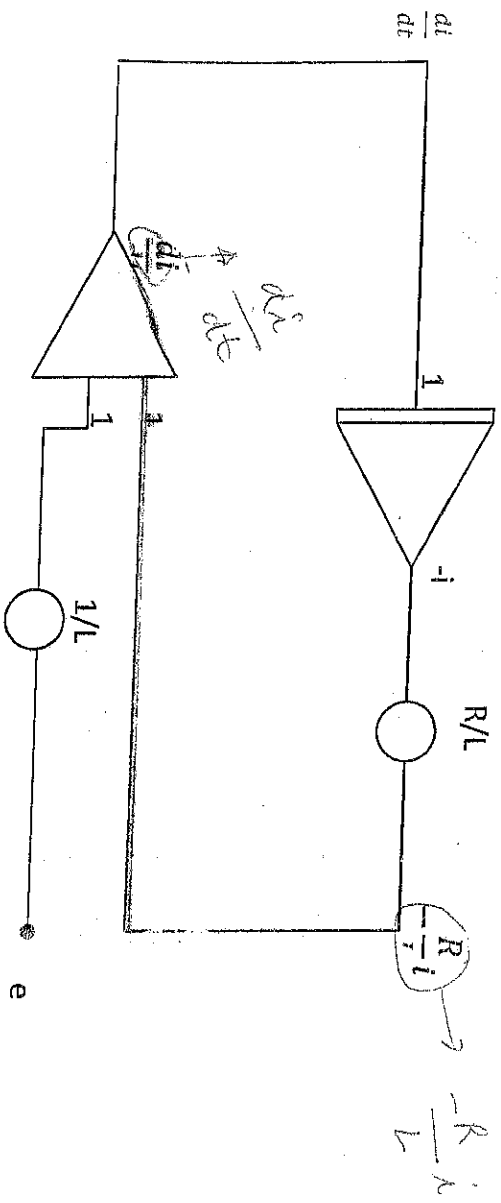


Fig. 5.5.2 Analogue computer flow diagram for R-L series circuit.

## R-L-C series circuit

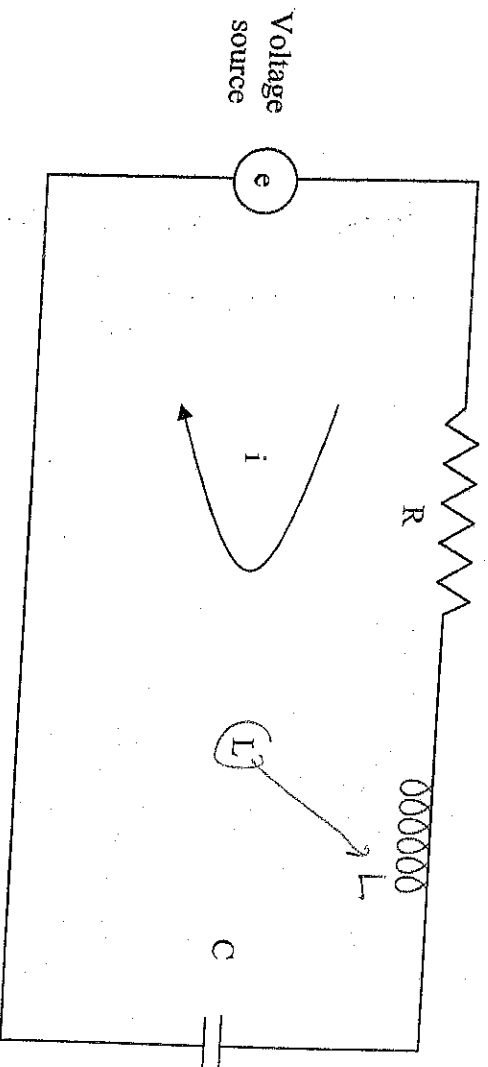


Fig. 5.5.2(b) series R-L-C circuit

The equation governing the system is

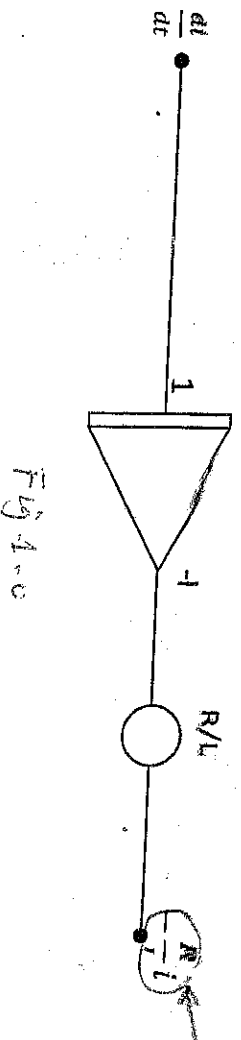
$$L \frac{di}{dt} + Ri + \frac{1}{C} \int_0^t i dt = e$$

$$L \frac{di}{dt} = e - Ri - \frac{1}{C} \int_0^t i dt$$

$$\frac{di}{dt} = \frac{e}{L} - \frac{Ri}{L} - \frac{1}{LC} \int_0^t i dt$$

The variables in the equation are  $\frac{di}{dt}$ ,  $i dt$ ,  $i$  and  $\int_0^t i dt$

The analogue diagrams that can represent the variables are realized as follows



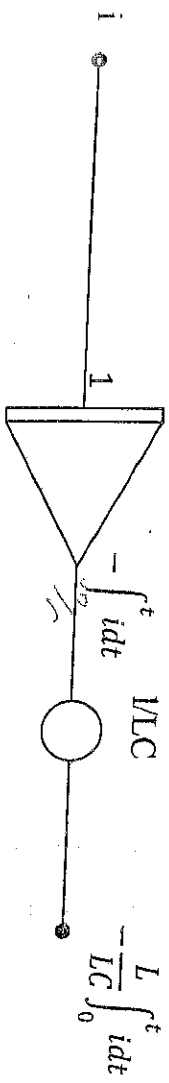


Fig. 2.0

Combining the three diagrams gives the analogue computer simulation of the system as follows in fig. 5.5.2(c)

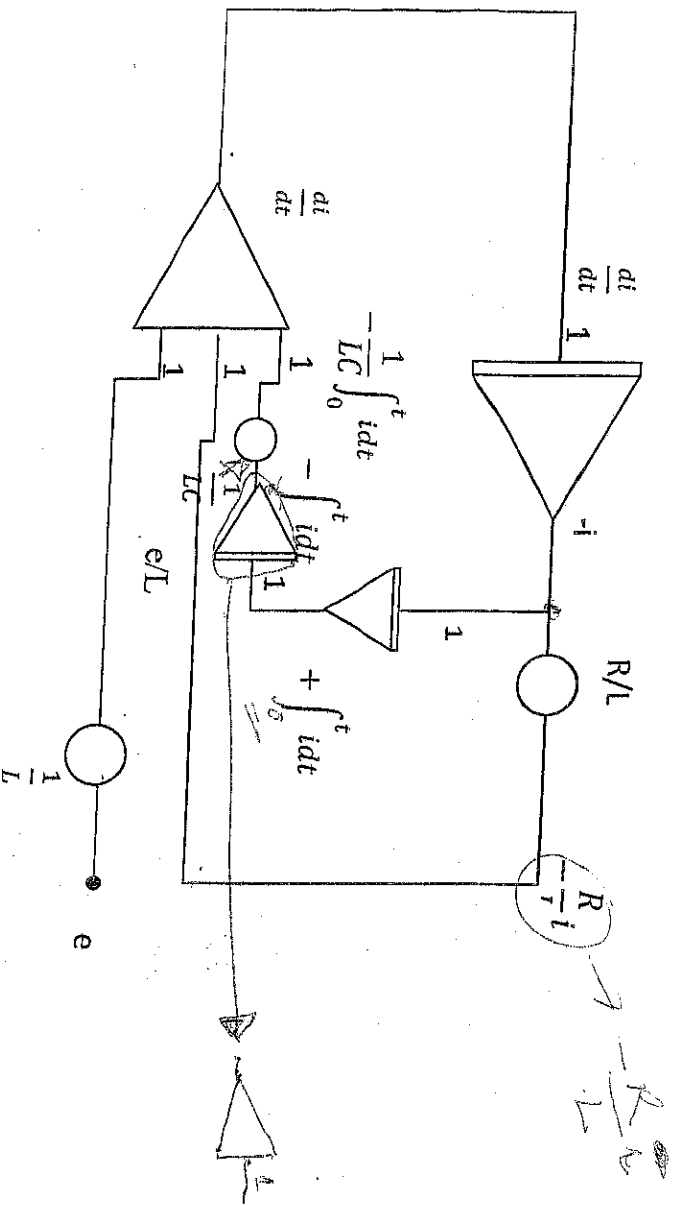


Fig. 5.5.2(c) Analogue computer flow diagram for series R-L-C circuit



### 5.5.3 Speed/position control systems

Field controlled d.c motor used for speed and position control system was discussed in section 3.3.1. The mathematical model of the motor was given as shown by the following equations.

$$J \frac{d\omega}{dt} + f\omega = T_m \quad \quad \quad 3$$

$$\text{and } \frac{J d^2\theta}{dt^2} + f \frac{d\theta}{dt} = T_m \quad \quad \quad 4$$

Equation (3) was used for speed control and equation (4) was used for position control.

#### (a) Speed control system

$$\frac{J d\omega}{dt} + f\omega = T_m \quad \therefore$$

$$\frac{d\omega}{dt} = \frac{T_m}{J} - \frac{f}{J} \omega$$

Or

$$\frac{J d\omega}{dt} = T_m - f\omega$$

where  $\omega$  = angular speed of motor

$T_m$  = torque developed by motor

The variables in this equation are  $\frac{d\omega}{dt}$  and  $\omega$  and can be represented with analogue diagram as follows in Fig. 1.6

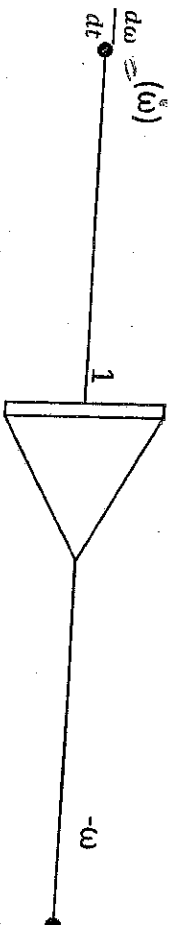
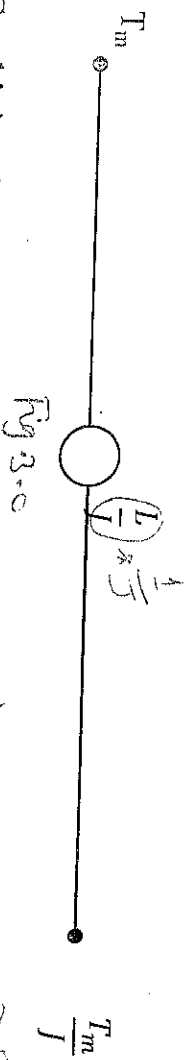


Fig. 1.6

The analogue diagram for  $-\frac{f}{J} \omega$  is



Also  $\frac{T_m}{J}$  is represented as follows



Combining the three diagrams gives the diagram of Fig. 5.5.3 (a).

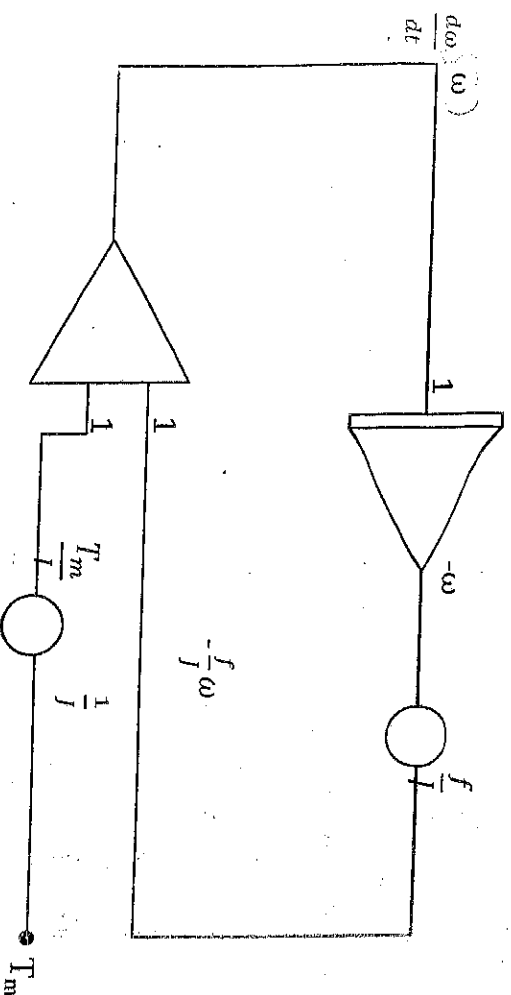


Fig. 5.5.3(a) Analogue computer flow diagram for speed control system with field controlled d.c. motor.

(b) Position control system

$$J \frac{d^2 \theta}{dt^2} + f \frac{d\theta}{dt} = T_m$$

Where  $\theta$  = angular position

$T_m$  = Torque developed by motor

$$\text{or } \frac{J d^2 \theta}{dt^2} = T_m - f \frac{d\theta}{dt}$$

$$\therefore \frac{J d^2 \theta}{dt^2} = \frac{T_m}{J} - \frac{f}{J} \frac{d\theta}{dt}$$

$$\text{Or } \ddot{\theta} = \frac{T_m}{J} - \frac{f}{J} \dot{\theta}$$

The variables in the above expressions are  $\theta$ ,  $\dot{\theta}$  and  $\ddot{\theta}$ .

The diagram is simulating the variables is

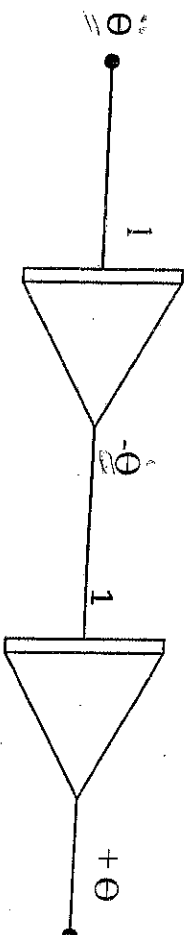


Fig. 1.0

The other symbols are

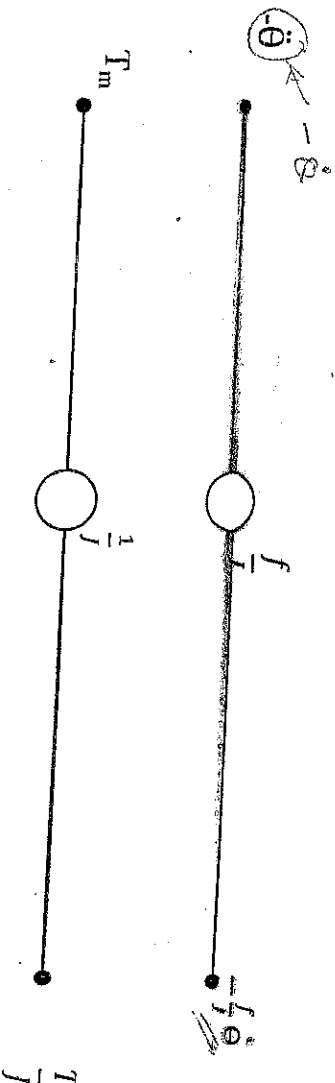


Fig. 2.0

Combining the diagram gives,

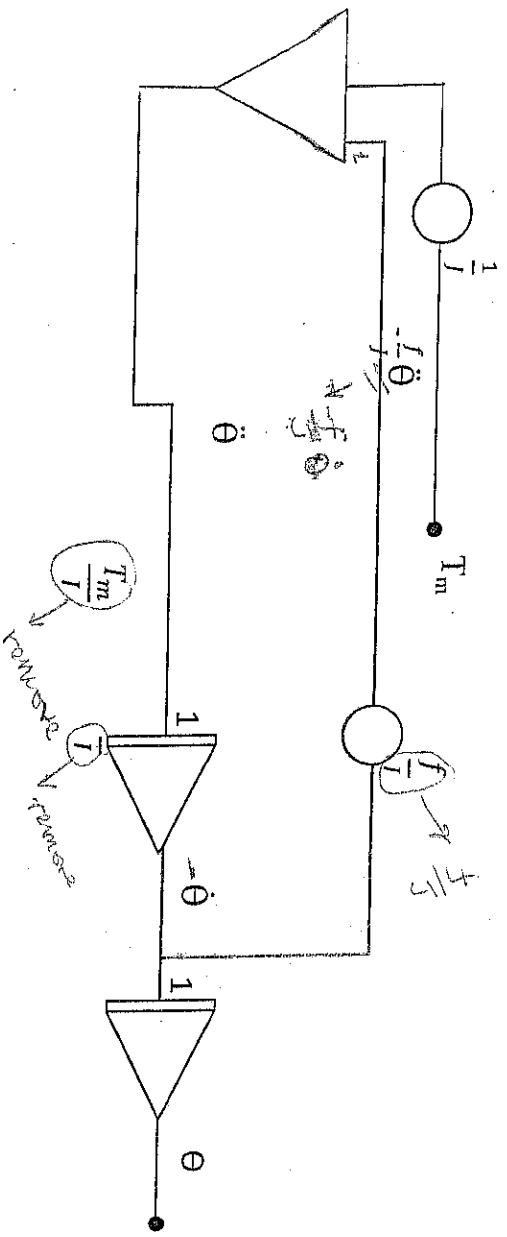


Fig. 5.5.3(b) Analogue computer flow diagram for position control system with field controlled d.c. motor

## 5.6 Scaling in analogue computing

If a process input variable  $x_1$  has a maximum value of say 1000 meters, then an amplifier whose output  $x_0$  would have to give output of 1000volts, which is impossible. Hence, the amplifier inputs and outputs cannot be numerically equal to the problem variable they represent, but merely proportional to them. We must multiply the problem variable by appropriate constants, called 'scale factors' to ensure that the amplifier's <sup>power</sup> inputs and output settings <sup>will</sup> ~~should~~ not be too large. In the <sup>vain</sup> amplifier inputs and output settings should not be too small.

Inputs and outputs settings will not be too large. In the same vain, amplifier inputs and outputs settings should not be too small.

A variable with maximum value of 0.002 meter cannot be accurately represented by an amplifier whose maximum output value is 0.002 volts <sup>value</sup> since this value is of the same order of magnitude as noise level in the computer which can introduce serious error. Hence, the need for scaling.

Similarly, it is important to assume that the rate of change of the computer variables is consistent with the dynamic properties of the computer, and that the solution takes place in a reasonable amount of time. Many physical processes take hours or days to complete, and others take only a few milliseconds or microseconds. To assume that the computer simulations takes place in a reasonable length of time, it is necessary to have some methods of speeding up or slowing down the solution, that is, a method of scaling the time as well as the amplifier.

### 5.6.1 Amplitude scaling

As pointed out above, the output of each amplifier should be proportional to the problem variable that it represents. The constant relating this two is called the scale factor. Scaling of the amplitude of variables is called 'amplitude scaling'. Amplitude scaling is defined by the relation;

$$\text{Amplifier output} = k \times (\text{problem variable})$$

$$\text{Or } k = \frac{\text{Amplifier output}}{\text{Problem variable}}$$

To determine an appropriate value for  $k$ , we need an estimate of the maximum value of the problem variable. Assuming this estimate is available, when the problem variable reaches its maximum value, then the amplifier output will be at its maximum value, which should be reference voltage of the computer.

$$k = \frac{\text{reference voltage}}{\text{maximum value of problem variable}}$$

Hence

$$\text{Amplifier output} = \frac{(\text{reference voltage})(\text{problem variable})}{\text{maximum value of problem variable}}$$

From this equation, it is obvious that when the problem variable is at its maximum, the amplifier output will be reference voltage, as it should be.

### Example

Suppose a problem variable  $x$  has a maximum value of 1000 meters; On a ten-volt computer, the scale factor would be,

$$K = \frac{10 \text{ volts}}{1000 \text{ meters}} = \underline{0.01 \text{ V/M}}$$

And the amplifier output would be

$0.1x$  or  $\frac{x}{10}$  be  $K = \frac{100 \text{ volts}}{1000 \text{ meters}} = 0.1 \text{ V/M}$ , and the amplifier output would be  $0.1x$  or  $\frac{x}{10}$ .

### 5.6.2 Time scaling

Scaling of time variable is called 'time scaling' it is customary to distinguish between time in the original problem and time on the computer by using the letters  $t$  and  $T$  respectively. Thus;

$t$  = 'problem time' = the independent variable in the original problem statement.

$T$  = 'machine or computer time' = the independent variable of the computer.

In time scaling, the computer variable is made proportional to the corresponding problem variable.

$$T = \beta t \quad (1)$$

Where  $\beta$  is the scale factor, <sup>solving</sup> scaling for  $\beta$

$$\beta = \frac{T}{t} = \frac{\text{computer time}}{\text{problem time}} \quad (2)$$

This  $\beta$  has the value of computer time (usually seconds) divided by problem time. If the problem time is also measured in seconds, then  $\beta$  is dimensionless. In this case, the magnitude of  $\beta$  indicates the factor by <sup>which</sup> what the problem is speeded up or slowed down. If  $\beta > 1$ , then the solution on the computer is slower than the original process; if  $\beta < 1$ , then the computer solution is faster than the original process. For example, if  $\beta = 5$ , then we see that  $T = 5t$ . when  $t = 1$  second,  $T = 5$  seconds, i.e. an event that takes place in one second in the original problem requires 5 seconds on the computer, the solution is slowed down by a factor of 5.

## 6.0 Need for scaling in analogue computing

The analogue computer is voltage limited to a practical range of operation of +100 volts as a maximum and 0 volt or -100-volts as a minimum. It is usually not possible to numerically equate the computer analogue voltages to the values of physical variables. Also, the analogue computer is limited as to the speed with which it will solve a problem. Practical bounds on solution time are 100 seconds as a maximum and 100 milliseconds as a minimum. Events in the physical world usually occur in time intervals which fall outside these limits.

Hence, the need to find a way to relate the amplitudes and times of the analogue computer to the magnitudes and times of the physical variable respectively. This is achieved through scaling. Scaling is relating the voltages and times of the analogue computer to the magnitudes and times of the physical variables respectively.

### 6.1 Amplitude scaling

The magnitudes or amplitudes of physical variables fall outside the analogue computer range and cannot be numerically equated to analogue computer outputs. The physical variables magnitudes must be made proportional to the analogue computer output voltage by scaling which makes problem variables amplitudes to

fall within the useful range of the machine. This procedure is called "Amplitude Scaling".

## 6.2 Choosing amplitude scale factors

As pointed out above, the output of each amplifier should be proportional to the problem variable that it represents. The constant relating the two is called the "scale factor" and is defined formally by the relation:

$$\text{Amplifier output} = k \times (\text{problem variable}) \quad (1)$$

$$\text{Where } k = \frac{\text{Amplifier output}}{\text{problem time}} \quad (2)$$

To determine an appropriate value for  $k$ , we need an estimate of the maximum value of the problem variable. Assume that this estimate is available. When problem variable reaches its maximum value, the amplifier output will be at its maximum value, which should be the reference voltage of the computer;

$$K = \frac{\text{Reference voltage}}{\text{maximum value of problem time}} \quad (3)$$

If we equate equation (2) and (3) we get,

$$\frac{\text{Amplifier output}}{\text{problem variable}} = \frac{\text{Reference voltage}}{\text{Maximum value of problem time}}$$

$$\therefore \text{Amplifier output} = \frac{(\text{Problem Variable})(\text{Reference voltage})}{\text{Maximum value of problem variable}} \quad (4)$$

From equation (4), it is obvious that when the problem variable is at its maximum, the amplifier output will be reference voltage, as it should be.

## 6.3 Example:

Suppose a problem variable  $x$  has a maximum value of 1000ft. On a ten volt computer, the scale factor would be from equation (2)



$$K = \frac{10 \text{ volts}}{1000ft} = 0.01 \text{ volt per foot}$$

And the amplifier output would be  $0.01x$  or  $\frac{x}{100}$

If a hundred volt computer were used, the scale factor would be;

$$K = \frac{100 \text{ volts}}{1000ft} = 0.1 \text{ volt per foot}$$

And the amplifier output would be  $0.1x$  or  $\frac{x}{10}$

The foregoing examples assumed that the voltage were measured in volts, but of course, we may use other units, such as microvolts, millivolts, or kilovolts, if these units turn out to be more convenient.

For the purpose of analogue scaling, the reference voltage of the computer is also referred to as 'analogue unit', 'machine unit', or 'normalized unit'.

#### 6.4 Scaled equation

When maximum values have been estimated and appropriate scale factors selected, the next step is to make the computer circuit consistent with the original problem equations. The original problem equations are stated in terms of problem variable, such as  $x, \dot{x}, \ddot{x}$ , etc, while the actual amplifier output are computer variables, such as  $\frac{x}{10}, \frac{\dot{x}}{10}, \frac{\ddot{x}}{10}$ , etc.

We must translate the original problem equations into computer oriented terms, i.e we must express each scaled amplifier output in terms of its scaled inputs.

An expression giving a scaled amplifier output in terms of the scaled inputs is called a scaled equation. A scaled equation can be written down for every amplifier in a problem.

## 6.5 Example

Suppose we have the following unscaled diagram of fig. 6.5

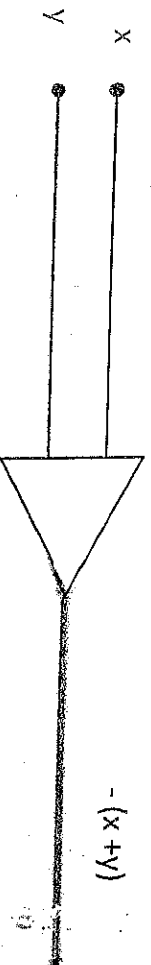


Fig. 6.5(a) *unscaled diagram of summer*

And assume we are given the following maximum values,

| Problem variable | Maximum value | Computer variable            |
|------------------|---------------|------------------------------|
| x                | 5 feet        | $\left[ \frac{x}{5} \right]$ |
| y                | 3 feet        | $\left[ \frac{y}{3} \right]$ |
| x + y            | ?             | ?                            |

If the maximum values are given for x and y, then the maximum value of the sum (x + y) may be easily estimated clearly, the sum can never be greater than 8 feet. Is this the true maximum value? Probably not, since this can only be achieved if x and y reach their maximum values in the same direction at the same time. If x is near zero when y is at its maximum, or if x is negative when y is positive, then the 8 feet level will never be achieved. However, if nothing is known about x and y except their maximum values, it makes good sense to be conservative and assume the “worst case”. Accordingly, we estimate the maximum value of the sum of (x + y) as 8 feet and the computer variable (scaled output) will be  $\left[ \frac{x+y}{8} \right]$

The relation between the output and inputs in unscaled form is simply;

$$-(x + y) = -(x + y) \quad (5)$$

Eliminating problem variables in favour of computer variables, we obtain;

$$8 \left[ - \left( \frac{x+y}{8} \right) \right] = - \left[ \frac{5}{8} \left( \frac{x}{5} \right) + 3 \left( \frac{y}{3} \right) \right] \quad (6)$$

Equation (6) is equivalent to equation (5), since we have substituted equivalent expressions. To solve for the amplifier output in terms of the inputs, we divide both sides of the equation by 8;

$$\left[ - \left( \frac{x+y}{8} \right) \right] = - \left[ \frac{1}{8} \left( \frac{5}{5} \right) \left( \frac{x}{5} \right) + \frac{1}{8} \left( \frac{3}{3} \right) \left( \frac{y}{3} \right) \right] \quad (7)$$

Equation (7) displays amplifier output, pot-settings, and gains. The scaled diagram corresponding to the equation is given in Fig. 6.5 (b) below

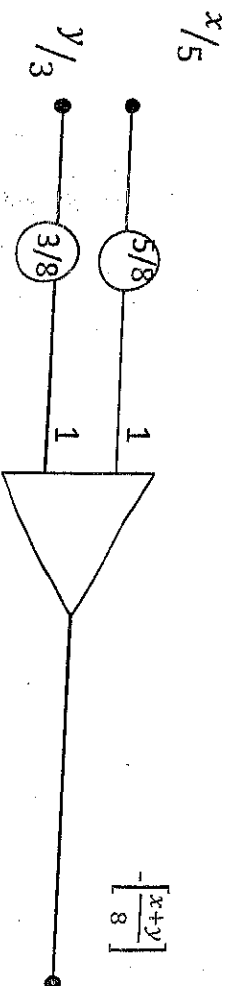
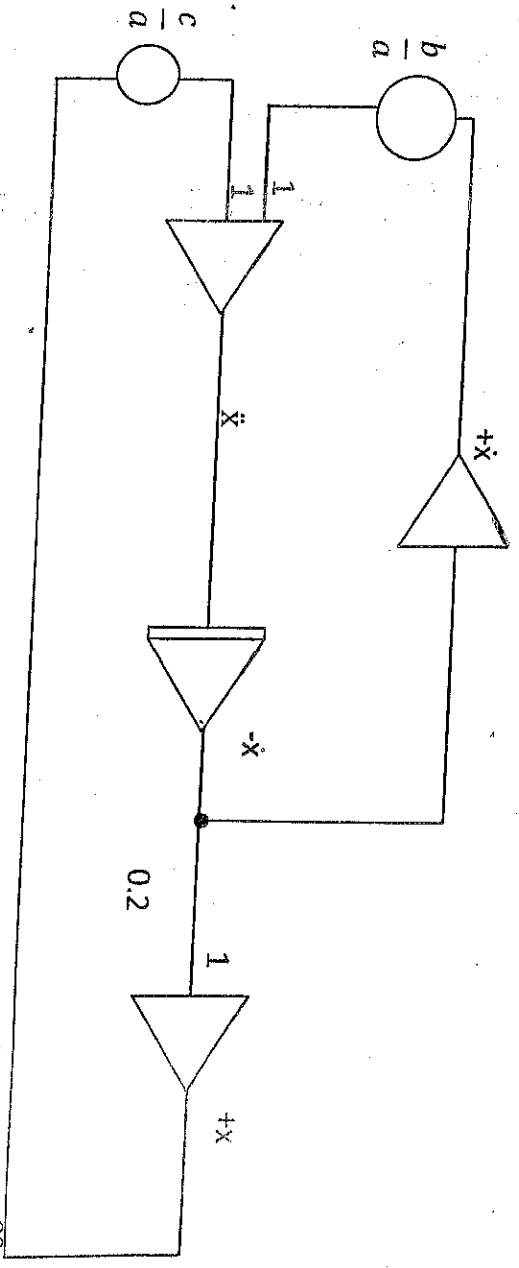


Fig. 6.5

## 6.6 Example

Consider the unscaled diagram if fig. 6.6 below;



*Fig. 6.6: Unscaled diagram of second order equation*

### 6.6.1 Scaling table for fig. 6.6

| Problem variable | Estimated value | Maximum | Computer variable                   |
|------------------|-----------------|---------|-------------------------------------|
| $x$              | 5               |         | $\left[\frac{x}{5}\right]$          |
| $\dot{x}$        | 10              |         | $\left[\frac{\dot{x}}{10}\right]$   |
| $-\ddot{x}$      | 10              |         | $\left[\frac{-\ddot{x}}{10}\right]$ |
| $\ddot{x}$       | 40              |         | $\left[\frac{\ddot{x}}{40}\right]$  |

The equation for the unscaled diagram is

$$\ddot{x} = -\frac{b}{a}\dot{x} - \frac{c}{a}x \quad (8)$$

substituting of computer variables into equation (8) gives:

$$40 \left[ \ddot{x}/40 \right] = - \left\{ 10 \frac{b}{a} \left[ \dot{x}/10 \right] + 5 \frac{c}{a} \left[ \frac{x}{5} \right] \right\} \quad (9)$$

Solving for  $\left[ \ddot{x}/40 \right]$ , we obtain;

$$\left[ \ddot{x}/40 \right] = - \left\{ \frac{10b}{40a} \left[ \dot{x}/10 \right] + \frac{5c}{40a} \left[ \frac{x}{5} \right] \right\}$$

$$\therefore \left[ \ddot{x}/40 \right] = - \left\{ \frac{b}{4a} \left[ \dot{x}/10 \right] + \frac{c}{8a} \left[ \frac{x}{5} \right] \right\} \quad (10)$$

Examining the factors in parenthesis indicate that they are less than unity, so that gains of 10 are required.

### 6.6.2 The integrator

(a) The unscaled equation for the unity rates producing  $x$  is

$$\frac{dx}{dt} = \dot{x} \quad (11)$$

Substituting computer variables;

$$5 \frac{d}{dt} \left[ \frac{x}{5} \right] = -10 \left[ \frac{-\dot{x}}{10} \right]$$

Solving for  $\frac{d}{dt} \left[ \frac{x}{5} \right]$ , we obtained *removed*

$$\frac{d}{dt} \left[ \frac{x}{5} \right] = \frac{-10}{5} \left[ \frac{-\dot{x}}{10} \right]$$

$$\therefore \frac{d}{dt} \left[ \frac{x}{5} \right] = -2 \left[ \frac{-\dot{x}}{10} \right] \quad (12)$$

Separating the factor on the right into the product of a pot-setting and a gain, we obtain;

$$\frac{d}{dt} \left[ \frac{x}{5} \right] = -10 (0.2) \left[ \frac{-\dot{x}}{10} \right] \quad (13)$$

The *initial* invited condition of this integrator is  $x_0$ , whose value may be assured to be equal to maximum value of  $x = 5$ , with pot set to  $\frac{x_0}{5}$ . Hence, the scaled equation for it is written as;

$$\begin{bmatrix} x_0/5 \end{bmatrix} = - \begin{bmatrix} -x_0/5 \end{bmatrix} (-1) \quad (14)$$

The equation shows that we need minus reference on the integrator pot to produce a positive on the integrator. As pointed out in the previous section, the integrator pot-setting has the scale factor as the amplifier output.

(b) The integrator producing  $-\dot{x}$  has the unscaled equation as

$$\frac{d\dot{x}}{dt} = \ddot{x} \quad (15)$$

Substituting computer variables;

$$10 \frac{d}{dt} \left[ -\dot{x}/10 \right] = -40 \left[ \ddot{x}/40 \right]$$

Solving  $\frac{d}{dt} \left[ -\dot{x}/10 \right]$ , we obtain;

$$\begin{aligned} \frac{d}{dt} \left[ -\dot{x}/10 \right] &= \frac{40}{10} \left[ \ddot{x}/10 \right] \\ \therefore \frac{d}{dt} \left[ -\dot{x}/10 \right] &= -4 \left[ -\ddot{x}/40 \right] \quad (16) \end{aligned}$$

Separating the factor on the right into product of a pot-setting and a gain we obtain;

$$\frac{d}{dt} \left[ \dot{x}/10 \right] = -10 (0.4) \left[ -\ddot{x}/40 \right] \quad (17)$$



achieving this or relating the computer time to physical time is called "Time Scaling".

Examination of the standard analogue computer components indicates that only one of them has anything to do with time – the integrator. Therefore, all components other than integrators are unaffected by time – scaling. If we examine the input/output relation for an integrator, we find that it integrates with respect to machine time "T", not problem time 't'.

## 7.1 Choosing time scale factors

If we want an output proportional to a variable x, the input should be proportional to  $\frac{dx}{dT}$ , rather than  $\frac{dx}{dt}$  since the problem is stated in terms of  $\frac{dx}{dt}$ , we find a way of converting one type of derivative into another.

It is customary to distinguish between time in the original problem and time on the computer by using the letters "t" and T respectively. thus

t = "problem time" = the independent variable in the original problem statement.

T = "machine time" or "computer time" = the independent variable on the computer.

In scaling, the computer variable is made proportional to the corresponding problem variable;

$$T = \beta t \quad (1)$$

Where  $\beta$  is the time-scale factor, <sup>solving</sup> scaling for  $\beta$



$$\beta = \frac{T}{t} = \frac{\text{computer time}}{\text{problem time}} \quad (2)$$

Thus  $\beta$  has the unit of computer time (usually seconds) divided by problem time. If problem time is also measured in seconds, the  $\beta$  is dimensionless. In this case, the magnitude of  $\beta$  indicates the factor by which the problem is speeded up or slowed down. If  $\beta > 1$ , then the solution on the computer is speeded up or slowed if  $\beta < 1$ , then, the computer solutions is faster than the original process; examples, if  $\beta = 5$ , then, from equation (1), we see that  $T = 5t$ . When  $t = 1$  second,  $T = 5$  seconds, i.e. an event that takes place in one second in the original problem requires 5 seconds on the computer, the solution is slowed down by a factor of 5.

From equation (1), it follows that:

$$\frac{dx}{dt} = \frac{1}{\beta} \frac{dx}{dt} \quad (3)$$

From equation (3), we see that if a voltage proportional to  $\frac{dx}{dt}$  is available as an input to an integrator, we need only multiply the integrator gain (or the pot-setting proceeding it) by  $\frac{1}{\beta}$  to convert from  $\frac{dx}{dt}$  to  $\frac{dx}{T}$  as shown in fig. 7.1

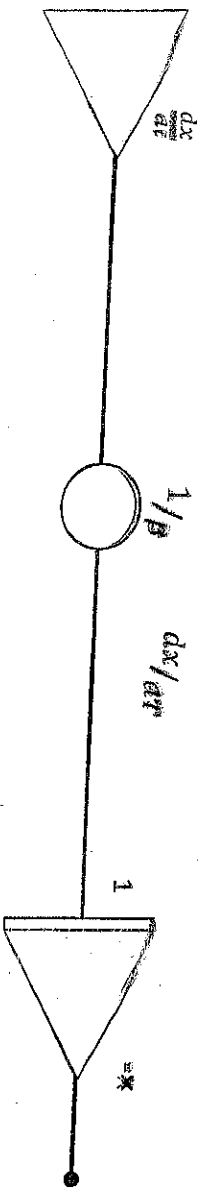


Fig. 7.1 conversion of  $\frac{dx}{dt}$  to  $\frac{dx}{T}$

Thus, we arrive at the general rule: "to speed up or slow down the computer solution by a factor  $\beta$ , every integrator input must be divided by  $\beta$ "

This factor  $\frac{1}{\beta}$  is in addition to the amplitude scale factor and other coefficients required by the problem.

## 7.2 Example

For example, suppose maximum value have been estimated for  $\dot{x}$  ( $=dx/dt$ ) and  $x$ . If  $\dot{x}$  is called as, say,  $[\frac{\dot{x}}{50}]$  and  $x$  is called as  $[\frac{x}{2}]$ , then we may write a scaled equation starting from equation (3) as follows,

$$\frac{dx}{dt} \cdot \frac{1}{\beta} \frac{dx}{dt} = \frac{1}{\beta} \dot{x} \quad (4)$$

Substituting the appropriate scaled variables as follows into equation (4), gives;

$$2 \frac{d}{dt} \left[ \frac{x}{2} \right] = 50 \cdot \frac{1}{\beta} \left[ \frac{\dot{x}}{50} \right]$$

Solving for  $\frac{d}{dt} \left[ \frac{x}{2} \right]$  which is the rate of change of the integrator we obtain;

$$\frac{d}{dt} \left[ \frac{x}{2} \right] = \frac{50}{2} \frac{1}{\beta} \left[ \frac{\dot{x}}{50} \right]$$

$$\therefore \frac{d}{dt} \left[ \frac{-x/2}{\beta} \right] = \frac{25}{\beta} \left[ \dot{x}/50 \right] \quad - (5)$$

If the sign of  $x$  and  $\dot{x}$  are as given in fig. 7.1, we must multiply by -1 to obtain;

$$\frac{d}{dt} \left[ \frac{-x/2}{\beta} \right] = - \frac{25}{\beta} \left[ \dot{x}/50 \right] \quad - (6)$$

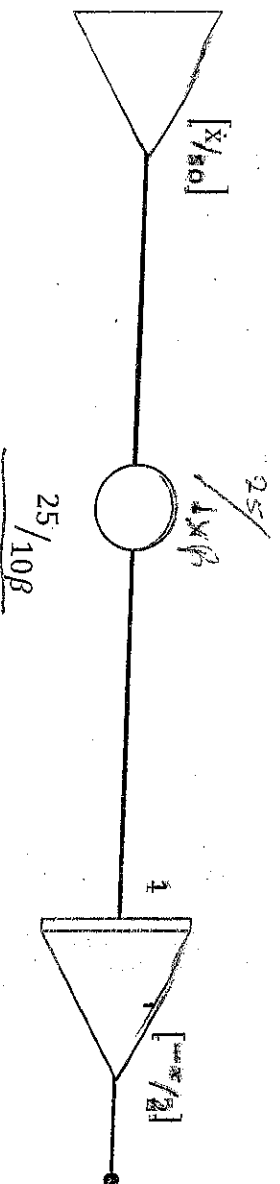
The factor  $\frac{25}{\beta}$  in parenthesis is the product of the pot-setting and integrator gain. If it is greater than one, we may use an appropriate input gain on the integrator. For example, if  $\beta = 100$ , then, the pot-setting  $\frac{25}{\beta}$  is  $\frac{25}{100}$  which is less than one. Assuming a gain of 1, then equation (6) can be written as equation (7) below;

$$\frac{d}{dt} \left[ \frac{-x/2}{\beta} \right] = -1 \left( \frac{25}{1x\beta} \right) \left[ \dot{x}/50 \right] \quad - (7)$$

But if  $\beta = 10$ , then, the pot-setting  $\frac{25}{\beta} = \frac{25}{10}$  which is greater than one but less than 10. Assuming gain of 10, we may rewrite the scaled equation as follows.

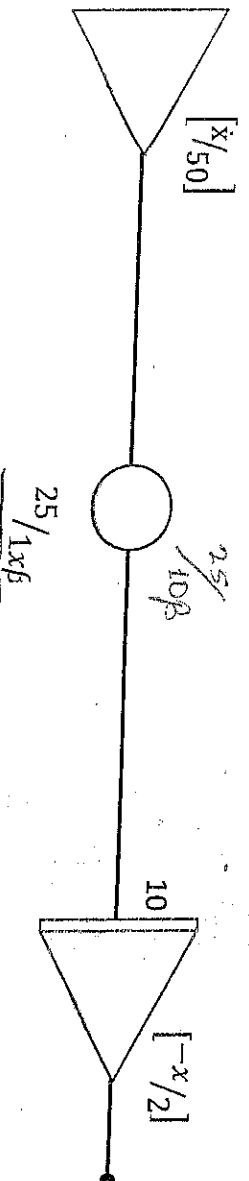
$$\frac{d}{dt} \left[ \frac{-x/2}{\beta} \right] = -10 \left( \frac{25}{10\beta} \right) \left[ \dot{x}/50 \right] \quad - (8)$$

For equation (7), the scaled diagram is given in fig. 7.2 (a) as follows



*Fig. 7.2(a): Scaled diagram of gain '1'*

For equation (8), the scaled diagram is given in fig 7.2(b) as follows;



*Fig. 7.2(b): Scaled diagram of gain '10'*

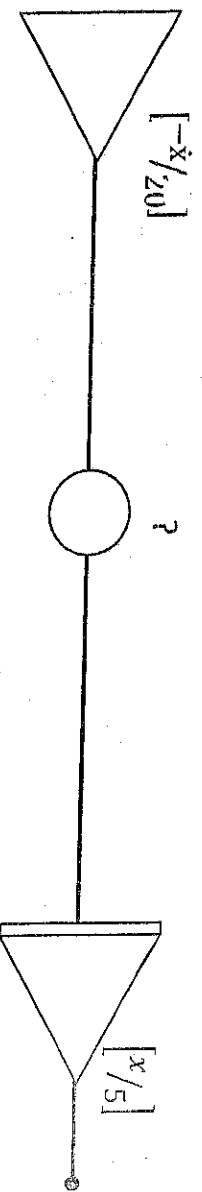
Note that pot-setting is labeled on the circuit diagram of fig. 7.2(b) as  $\frac{25}{10\beta}$  (or it may be reduced to  $\frac{5}{2\beta}$ ).

Merely writing the numerical value (0.2500) on the circuit would not reveal the time-scale factor explicitly.

The magnitude of an integrator input gain determines the rate of change of its output, “large integrator input gains indicate a fast solution and small integrator input gains indicate a slow solution.” Note that the same time-scale factor must be used for every integrator.

### 7.3 Short-cuts in time-scaling

For example, consider the portion of a circuit diagram given in Fig. 7.3(a) below;

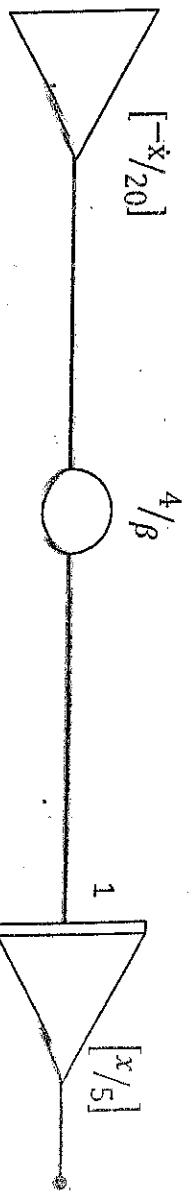


*Fig. 7.3(a): For unknown time-scale?*

The problem is to calculate the value of the pot-setting necessary to reconcile the scale factor of  $x$  with the scale factor  $\dot{x}$ . A “common-sense” rule is that the pot-setting must contain the “rates of the two scale factors”. This rule might be stated informally as follows “cancel the incoming scale factor and introduce the desired output scale factor”. In the diagram of fig. 7.3(a), it is fairly obvious that we need to multiply the input by 20 (to cancel the incoming factor of  $\frac{1}{20}$ ) and divide by 5 (to introduce the desired scale factor of  $\frac{1}{5}$  at the output). Hence, the product of the pot-setting and the gain should be  $\frac{20}{5\beta} = \frac{4}{\beta}$ , including the time-scale factor. The scaled

$$\text{equation for fig. 7.3(a) is } \frac{d}{dt} \left[ \frac{x}{5} \right] = -1 \left( \frac{4}{10\beta} \right) \left[ -\dot{x}/20 \right] \text{ --- (9)}$$

with scaled diagram as shown in fig. 7.3(b).

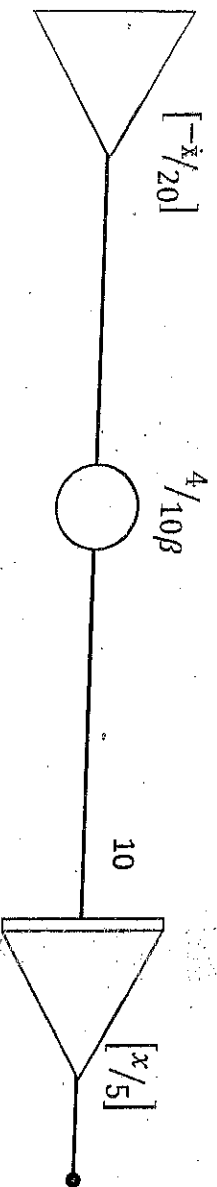


**Fig. 7.3(b): Scaled diagram of fig. 7.3(a) with gain of "1".**

Note the "product" of the pot-setting and gain in equations (9) and (10). Another form of scaled equation for fig. 7.3(a) is shown below;

$$\frac{d}{dt} \left[ \frac{\dot{x}}{5} \right] = -10 \left( \frac{4}{1x\beta} \right) \left[ \frac{-\dot{x}}{20} \right] \quad (10)$$

And the scaled diagram is shown in fig. 7.3(c),



**Fig. 7.3(c): Scaled diagram of fig. 7.3(a) with gain of "10".**

Fig. 7.3(b) and fig. 7.3(c) are the same and give same results. From fig. 7.3 (b) and fig. 7.3(c), it implies that any number that is used to divide pot-setting must also be used to multiply the input gain of an integrator to maintain the balance.